

Semester DB Project: Phase III

Programming Language Reference Database

CSC 436 | Matt Petrarca, Matt Barbrack, Ryan Megna, Victor Paulino, Hudson Byers

Part I: Review Feedback & Revision Plan

Strengths Identified

The following strengths were highlighted by peers during the Database Expo:

- Scalable database design that allows clean additions of new languages and functions.
- Strong visual design that maintains a consistent programming and terminal aesthetic.
- Clear and thorough presentation covering security, optimization, and application features.
- The CLI lookup interface was praised as a useful and well executed feature.
- 3NF compliance was specifically called out as a strength, noting how it removed redundancy and collision issues.
- Offering both a terminalstyle CLI and a web interface gave users multiple ways to interact with the data.

Weaknesses and Suggested Improvements

The following concerns and suggestions were collected from peer feedback forms:

- Multiple reviewers noted the green on black color scheme was difficult to read, with comments about eye strain and text that was too small.
- The reviewer requested a light mode option.
- Reviewers wanted to see a live demo of language lookup comparisons during the presentation.
- Some peers questioned how the database would be efficiently populated as more languages are added over time.
- A suggestion was made to make the project open source.
- One reviewer noted it would be helpful to have a centrally hosted database that users do not need to manage themselves.

Revision Plan

After evaluating the feedback as a group, we determined that the most impactful and feasible revisions before final submission were focused on UI readability and accessibility. Feedback on text size and contrast appeared consistently across multiple

reviewers and represented a direct usability problem that could be addressed through targeted CSS changes. Suggestions such as user accounts or open sourcing the project would require significant new development and fall outside the scope of this phase.

The specific revisions planned were:

- Update the color scheme across all pages to improve contrast and reduce eye strain while keeping the terminal aesthetic.
- Increase the base font size from 13px to 17px across the public interface and all admin pages.
- Add a light mode toggle to the public facing page and the admin dashboard.
- Replace all hardcoded nearinvisible colors throughout labels, footers, panel headers, and card text with CSS variables that respond correctly to both dark and light themes.

Part II: Implementation of Revisions

The revisions identified in Part I were implemented in the PHP source files for the public interface (index.php), the admin dashboard (admin/index.php), and the admin login page (admin/login.php). All changes were made to the CSS layer only and do not affect any database queries, schema, or server side logic. PHP, SQL, and JavaScript functionality remained unchanged.

Changes were applied across the following three files:

- index.php: the public facing browse and compare interface
- admin/index.php: the admin dashboard used to insert records
- admin/login.php: the admin authentication page

Part III: Documentation

Revision 1: Color Scheme and Contrast

What was changed: The CSS color variables across all three pages were updated. The background was changed from pure black (#000000) to a dark navy (#0d1117). The primary green was softened from neon (#00ff88) to a more natural shade (#3fb950). Body text was brightened from #c8c8c8 to #e6edf3 for better contrast.

Why it was changed: Multiple peer reviewers specifically called out the greenonblack combination as hard to read. Pure black backgrounds paired with neon colors produce

glare that causes eye fatigue over time. The updated palette preserves the terminal aesthetic while being significantly more comfortable to use.

How it improved the project: The interface is now easier to read for extended use without losing its visual identity. All text, labels, and interactive elements remain distinguishable and the overall look of the application is unchanged.

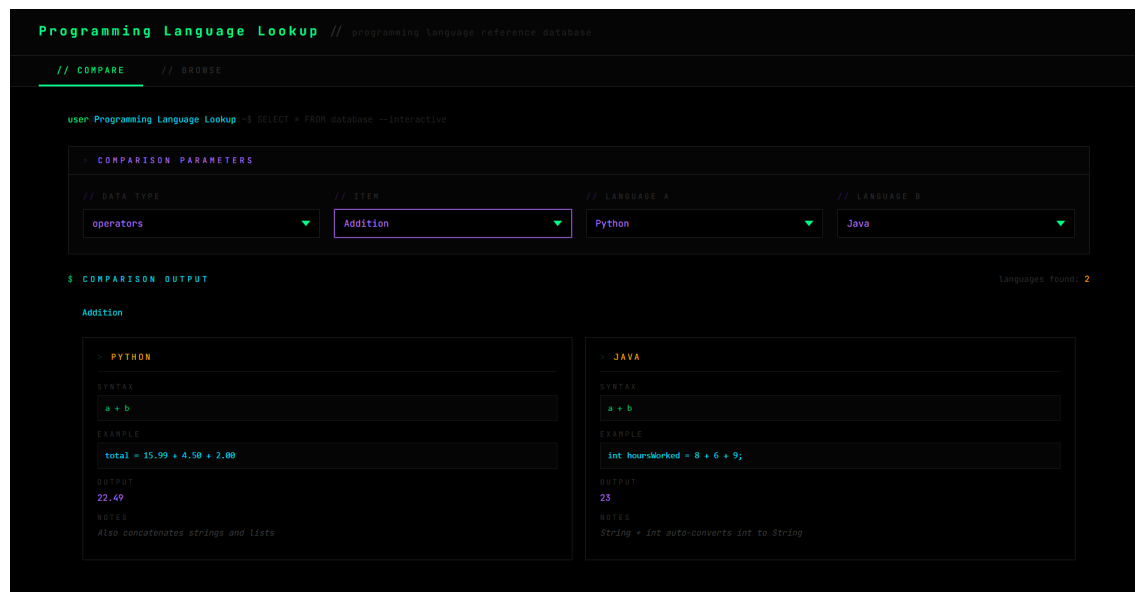
Revision 2: Font Size Increase

What was changed: The base font size across all three pages was increased from 13px to 17px. Individual element sizes including table cells, dropdown labels, panel titles, tab buttons, and card text were also increased proportionally.

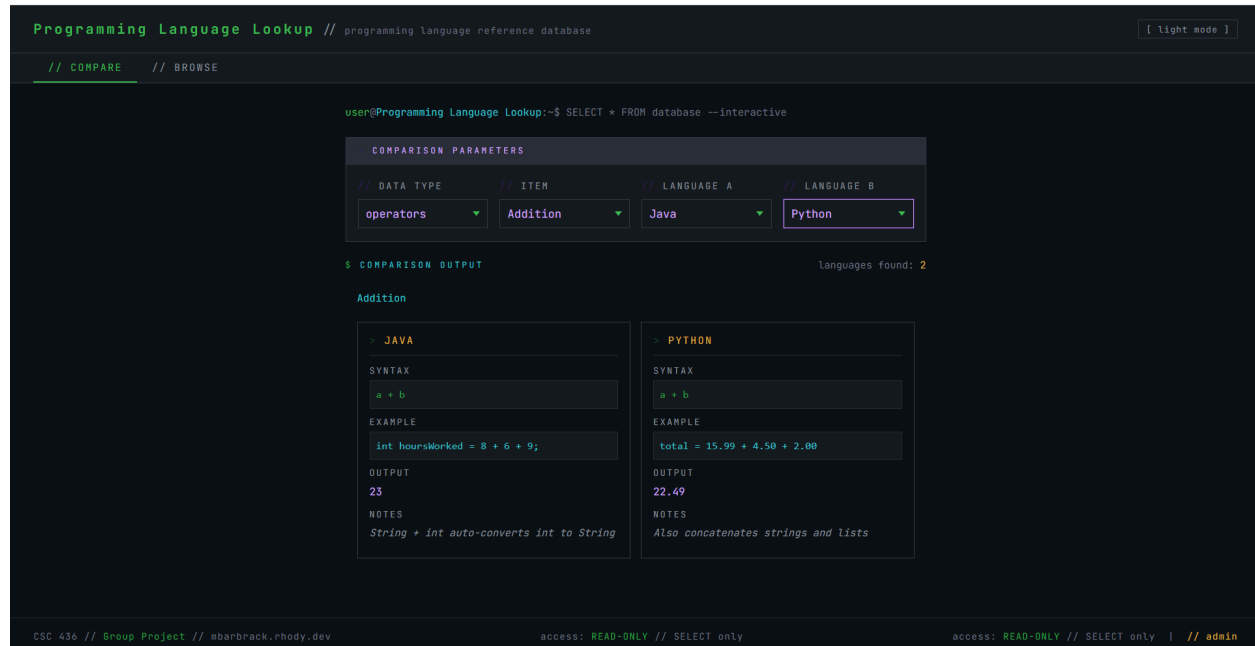
Why it was changed: Two reviewers specifically requested larger text, with one noting they could not read the interface at its current size. A 13px base is below comfortable reading standards for web content, especially for users with any degree of visual impairment.

How it improved the project: The interface is now comfortably readable without straining. Because all sizing used relative rem units, the single base font change scaled the entire interface consistently without breaking the layout.

Original dark interface before color changes/Interface with original 13px font size



New dark interface after color changes/Interface after font size increase to 17px



Revision 3: Light Mode Toggle

What was changed: A light mode toggle button was added to the header of the public interface and the admin dashboard. Clicking it applies a `.light` class to the body element which activates a complete set of CSS variable overrides, switching to a white background with dark text and adjusted accent colors.

Why it was changed: Light mode was directly requested by multiple reviewers. Some users prefer reading dark text on a light background, particularly in bright environments. Offering both modes makes the application accessible to a wider range of users.

How it improved the project: Users can now switch between dark and light mode instantly without a page reload. Both modes are fully styled across the header, tabs, panels, dropdowns, tables, compare cards, and footer on both the public and admin pages.

Public interface in light mode

Programming Language Lookup // programming language reference database [light mode]

// COMPARE // BROWSE

```
user@Programming Language Lookup:~$ SELECT * FROM database --interactive
```

> COMPARISON PARAMETERS

// DATA TYPE	// ITEM	// LANGUAGE A	// LANGUAGE B
operators	Addition	Java	Python

\$ COMPARISON OUTPUT Languages found: 2

Addition

> JAVA

SYNTAX

a + b

EXAMPLE

int hoursWorked = 8 + 6 + 9;

OUTPUT

23

NOTES

String + int auto-converts int to String

> PYTHON

SYNTAX

a + b

EXAMPLE

total = 15.99 + 4.50 + 2.00

OUTPUT

22.49

NOTES

Also concatenates strings and Lists

CSC 436 // Group Project // mbanbrack.rhody.dev access: READ-ONLY // SELECT only access: READ-ONLY // SELECT only // admin

Admin dashboard in light mode

Programming Language Lookup // ADMIN user: admin [light mode] [logout]

// database management

// LANGUAGE // FUNCTION // OPERATOR // DATA STRUCTURE // IMPLEMENTATION

```
admin@Programming Language Lookup:~$ INSERT INTO database --interactive
```

ADD IMPLEMENTATION

// LANGUAGE *

select language...

// SYNTAX

e.g. print(value)

// RESULT / OUTPUT

e.g. Hello World

// EXAMPLE

e.g. print('Hello', 42)

// NOTES

e.g. Supports multiple arguments

LINK TO EXISTING ITEM *

FUNCTION OPERATOR DATA STRUCTURE

select item...

Links this implementation to a function / operator / structure via the junction

Revision 4: Hardcoded Color Fixes

What was changed: Dozens of hardcoded color values such as #2a2a2a, #333, #444, #050505, and #030303 used throughout labels, panel backgrounds, footers, card notes, prompt lines, and the compare output section were replaced with CSS variables.

Why it was changed: These hardcoded dark colors were nearly invisible against the dark background and completely failed to adapt when light mode was toggled. Labels, helper text, footer content, and placeholder messages were unreadable.

How it improved the project: All previously invisible or nearinvisible text is now clearly readable in both modes. Every element correctly responds to the theme toggle because colors are now defined through CSS variables rather than hardcoded values.